Intelligent Rail Spine: Jing-Zhang Herringbone · Centennial AI Innovation Belt

Turning railway heritage from a backdrop into an accessible, interpretable, exit-ready AI public-service spine.

This booklet supports independent review and condenses the spatial, service, risk, and implementation logic in the V2 technical package. It is not a regulatory plan, construction drawing, implementation approval, government commitment, field measurement, or deployment statement. All geometry retains the provisional boundary; the full package must be recomputed when formal data are released.

One belt, two wings, three areas: distinct urban roles joined by one rule

The overall design scope, three key areas, and two wings are not parallel layers. They are linked by a heritage–service–governance rule: the spine carries continuous public space; Zhongzhiyuan supports governance and full-stack innovation; Origin Community supports near-campus transfer and talent services; Bell-Hub supports station-city exchange and international presentation.

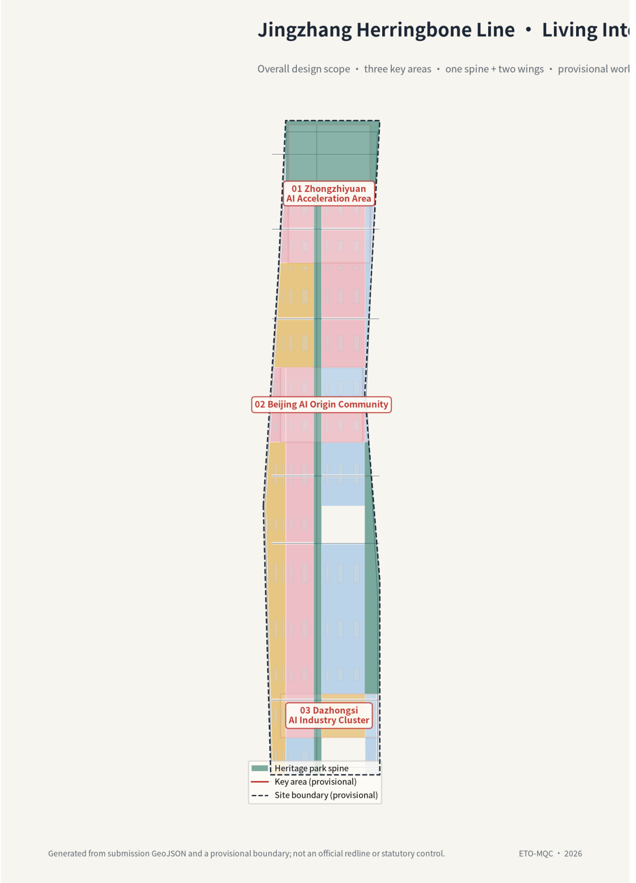

Figure evidence anchor

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Provisional boundary · Three scope levels · Space–service–governance

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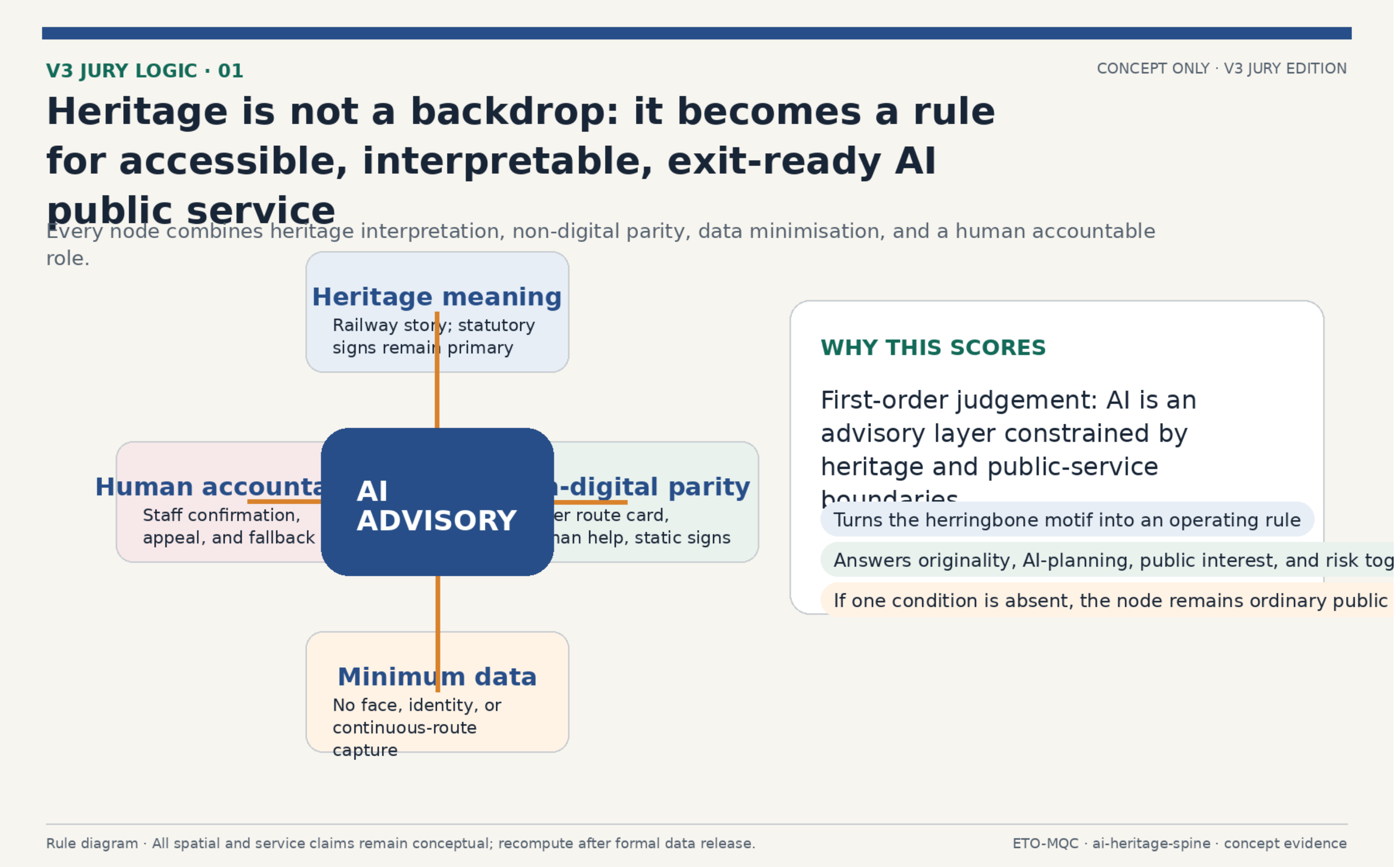
Every node passes four public-service conditions together

The herringbone moves from a visual motif to an operating rule. If a node cannot jointly deliver heritage meaning, non-digital parity, data minimisation, and a human accountable role, it cannot be described as an AI public-service node; it remains ordinary public space.



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Originality: rule, not slogan | AI planning: bounded | Public interest: exit-ready

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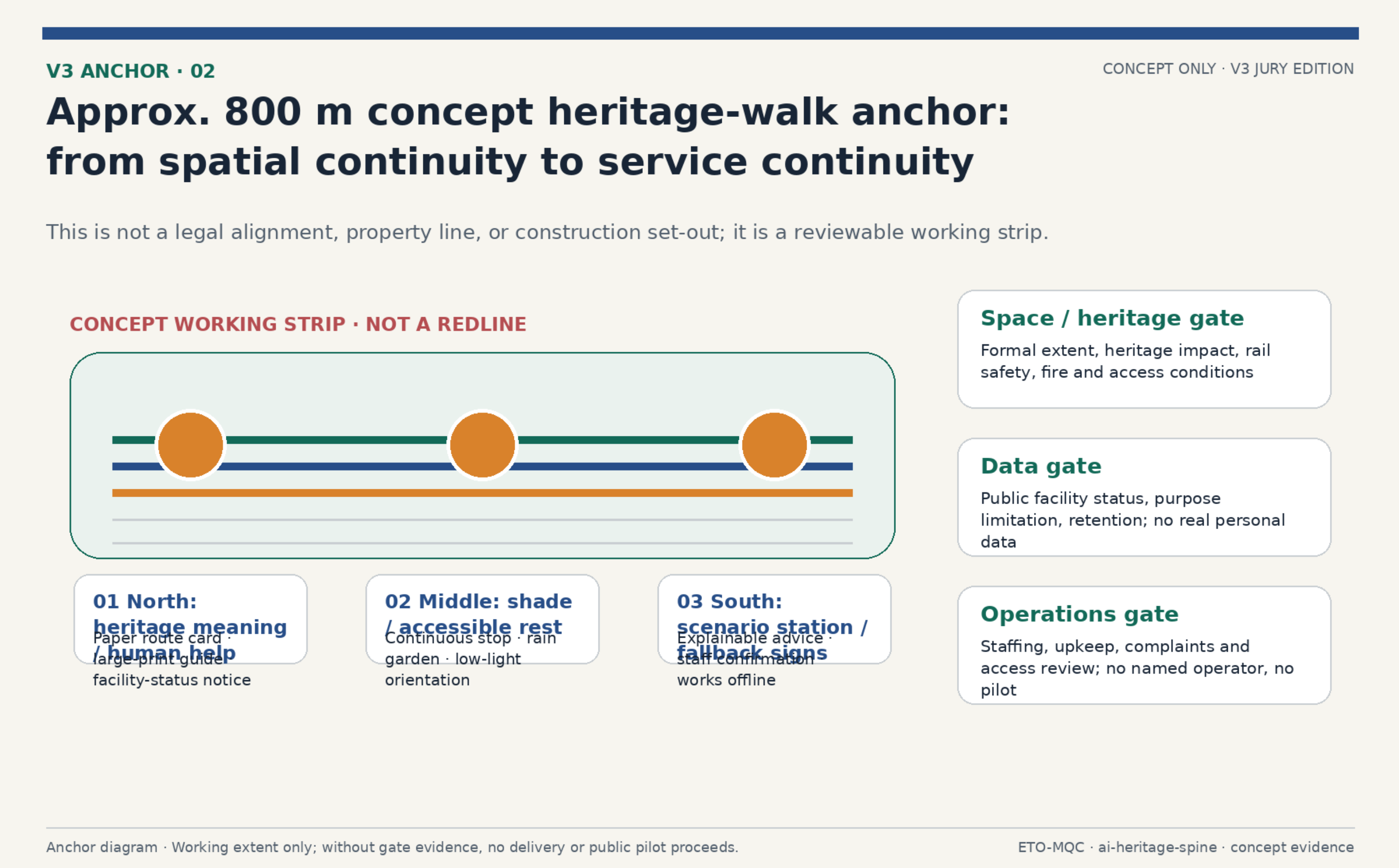
An approximate 800 m concept heritage-walk working strip

V3 selects an approximate 800 m concept working strip within the spine to show a minimum loop: north heritage interpretation/human help, middle shade/accessible rest, and south scenario station/fallback signs. It is not a statutory alignment, property line, or construction set-out.



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Space/heritage gate · Data gate · Operations gate · Public gate

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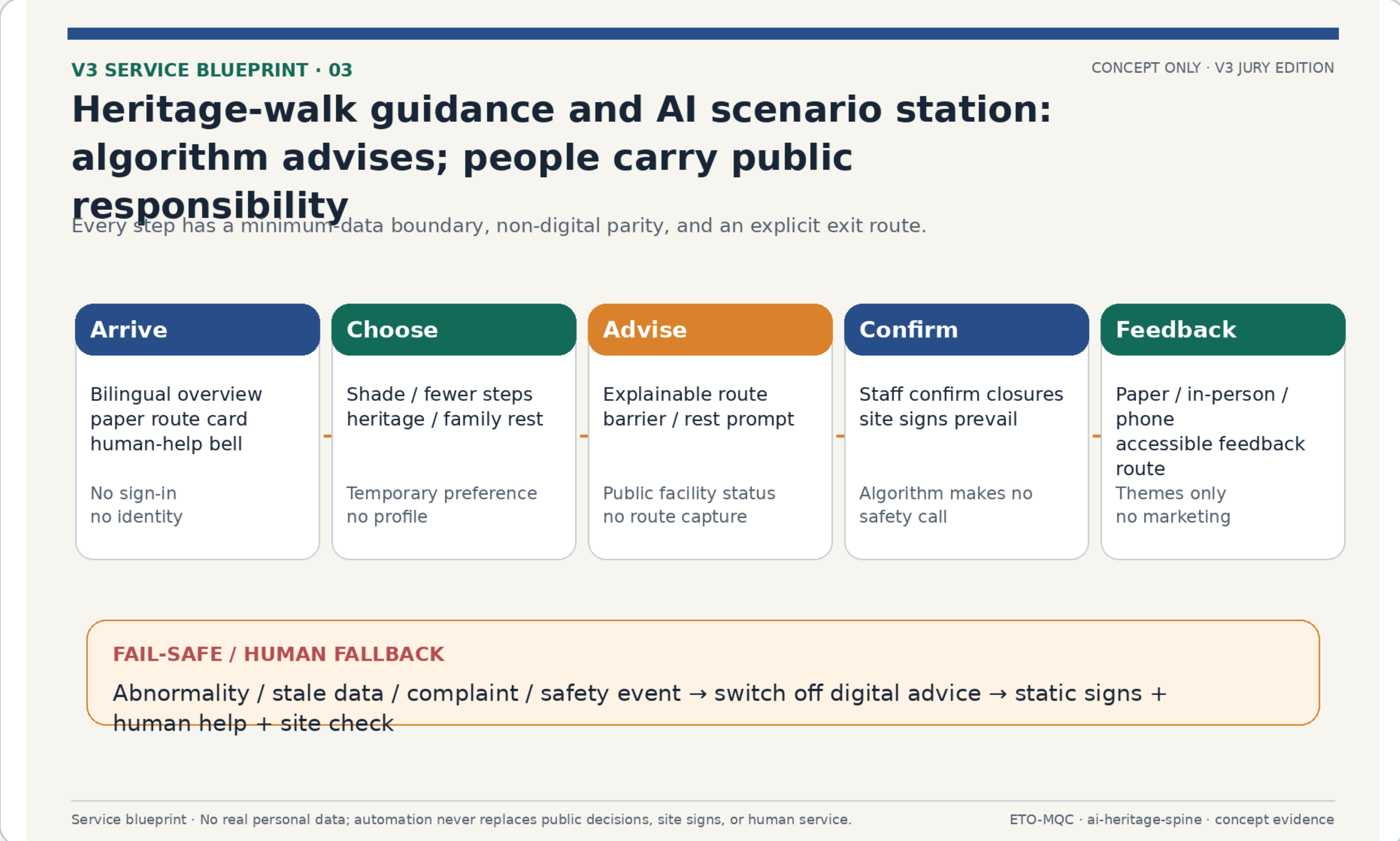
Scenario station: algorithm advises; people carry public responsibility

The primary scenario reframes walk guidance as a low-risk orientation and access-service prototype. It requires no login and captures no face, identity, or continuous route; staff and site signs prevail over algorithms. Stale data, complaints, or safety incidents turn off digital advice and restore static signs, human help, and site checks.



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Minimum data | Human confirmation | Static fallback

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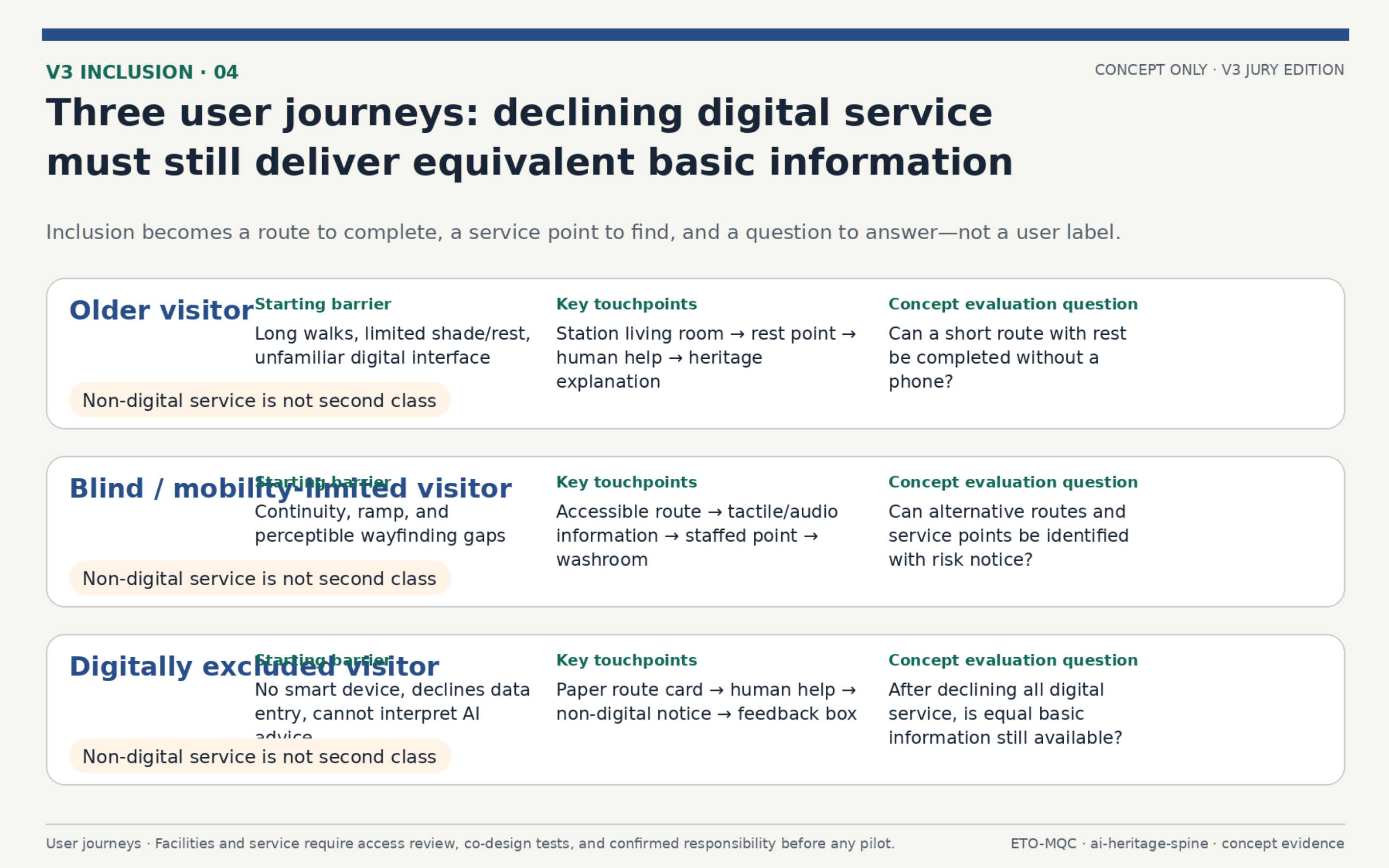
Three user journeys: a complete service remains possible without a phone

Older visitors, blind or mobility-limited visitors, and digitally excluded visitors are represented through starting barriers, touchpoints, answerable review questions, and fallback safeguards. The core test is simple: after declining all digital service, does the visitor still receive equivalent basic information?



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All-age friendliness | Accessibility | Paper and human service in parallel

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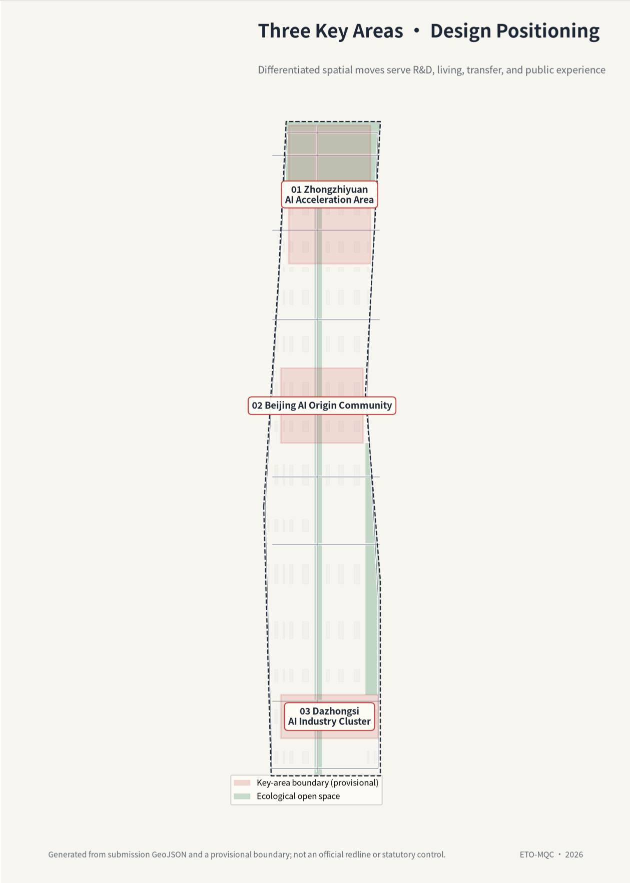
Zhongzhiyuan, Origin Community, and Bell-Hub: roles—not three copies of an AI park

Zhongzhiyuan hosts full-stack innovation and governance display; Origin Community hosts near-campus transfer and talent services; Bell-Hub hosts station-city exchange and international presentation. Each area has distinct spatial action, public benefit, and implementation dependencies.



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Governance and evaluation | Transfer and talent | Exchange and presentation

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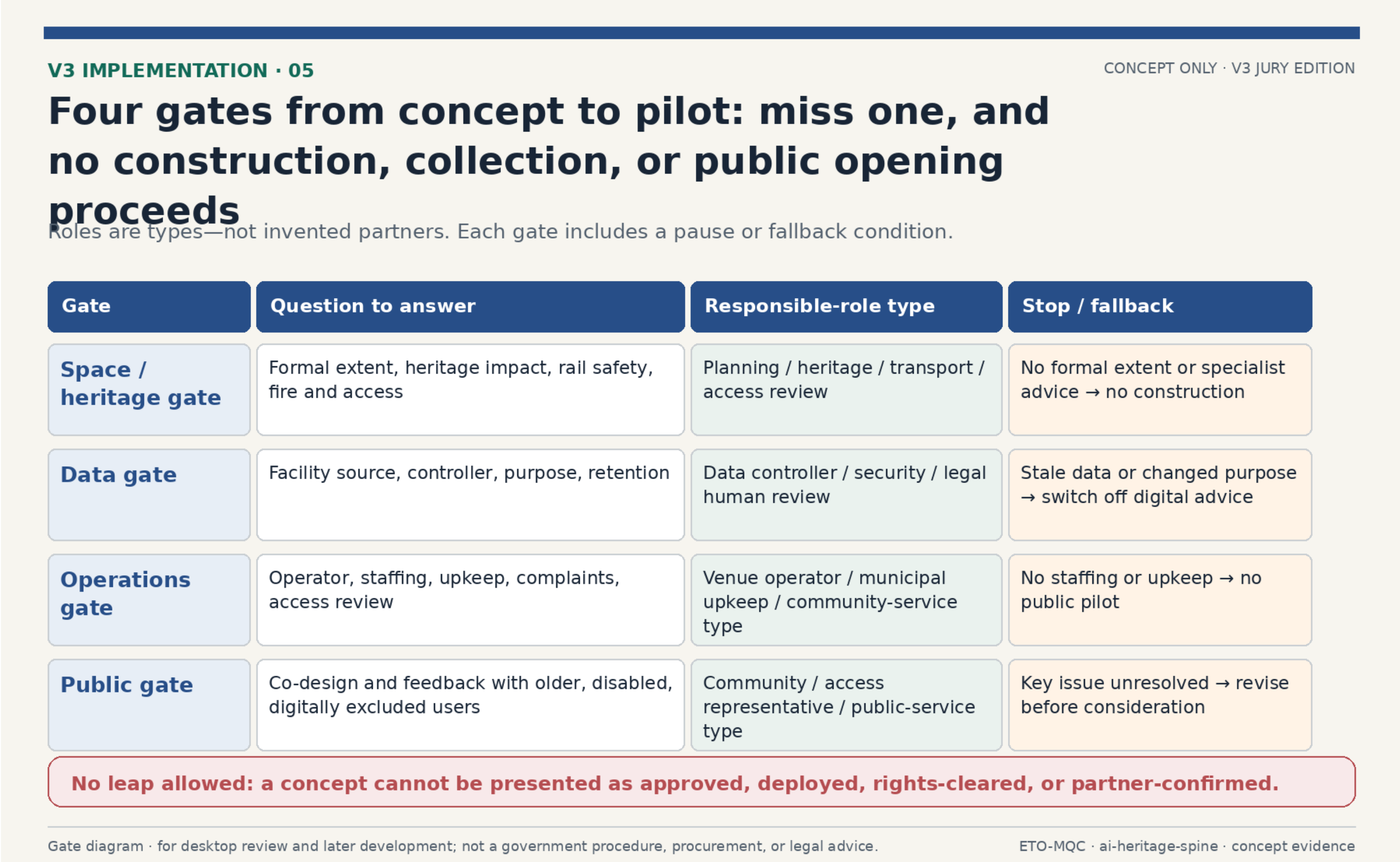
From concept to pilot: miss one gate and no construction, collection, or public opening proceeds

Implementation responsibilities are expressed as role types, not invented institutions. Without a formal extent, heritage/safety advice, data control and purpose limitation, staffing and upkeep, or public co-design testing, the concept cannot proceed to construction, real-data collection, or a public pilot.



V3 IMPLEMENTATION · 05

CONCEPT ONLY · V3 JURY EDITION

Four gates from concept to pilot: miss one, and no construction, collection, or public opening proceeds

Roles are types—not invented partners. Each gate includes a pause or fallback condition.

Gate	Question to answer	Responsible-role type	Stop / fallback
Space / heritage gate	Formal extent, heritage impact, rail safety, fire and access	Planning / heritage / transport / access review	No formal extent or specialist advice → no construction
Data gate	Facility source, controller, purpose, retention	Data controller / security / legal human review	Stale data or changed purpose → switch off digital advice
Operations gate	Operator, staffing, upkeep, complaints, access review	Venue operator / municipal upkeep / community-service type	No staffing or upkeep → no public pilot
Public gate	Co-design and feedback with older, disabled, digitally excluded users	Community / access representative / public-service type	Key issue unresolved → revise before consideration

No leap allowed: a concept cannot be presented as approved, deployed, rights-cleared, or partner-confirmed.

Gate diagram · for desktop review and later development; not a government procedure, procurement, or legal advice.

ETO-MQC · ai-heritage-spine · concept evidence

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Pause first

Human review

No presumed partners

From concept to pilot: miss one gate and no construction, collection, or public opening proceeds

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Gate	Required condition	Stop/fallback
Space/heritage	Formal extent; heritage, rail, fire, and access conditions	No formal extent or advice: no construction
Data	Source, purpose, controller, retention	Stale data or changed purpose: switch off digital advice
Operations/ public	Staffing, upkeep, complaints, co-design, access review	No responsibility mechanism or unresolved issue: no pilot

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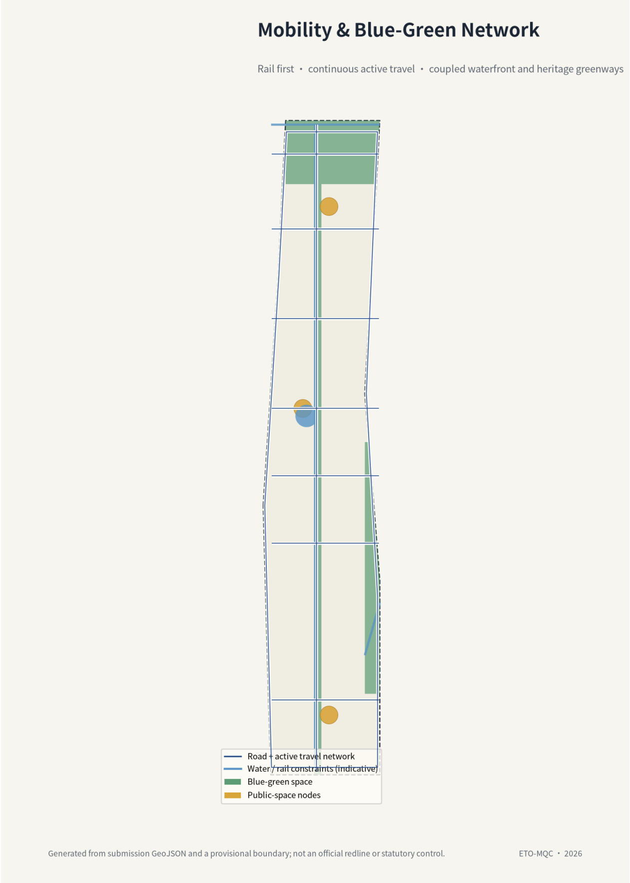
Not linear "delivery", but pausable, reviewable, and exit-ready

Stage one fills evidence on boundaries, tenure, heritage, safety, accessibility, and data. Stage two uses limited desktop simulation and co-design before any further professional scoping. Stage three considers expansion only when public benefit, safety, legality, and operational accountability are confirmed. Any failure restores non-digital service or stops.



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Evidence gap closure · Co-design · Review / exit

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Spatial facts sit beside professional unknowns

Site, key-area, green-space, public-space, and road metrics come from V2 layers recomputed in EPSG:4548. FAR, building height, density, tenure, statutory road lines, and field-service performance remain unknown. Unknown is not a missing page; it is a professional boundary.

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EPSG:4548 | Recompute after formal data | Vision cannot fill unknowns

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11.41 km² Overall scope; provisional-boundary calculation	**369.3 ha** Three key areas; provisional polygons	**19.76%** Green-space share; green_space/site
1.13% Public-space share; public_space/site	**39.36 km** Road centrelines; roads-layer calculation	**12/5/3/8** Scenarios/personas/ landmarks/projects

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A shared boundary for annual events, community, and VI: public issues first, rights-clear, auditable openness

AI Week, the developer community, scenario opening, and brand assets are operating suggestions only. They require open rules, accessibility, conduct standards, copyright/logo permissions, purpose-limited data, and withdrawal mechanisms. They are not confirmed events, subsidies, admission rights, or public partnerships.

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Accessibility | Rights ledger | Open but exit-ready

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The next step is evidence, co-design, and review—not calling a concept implementation

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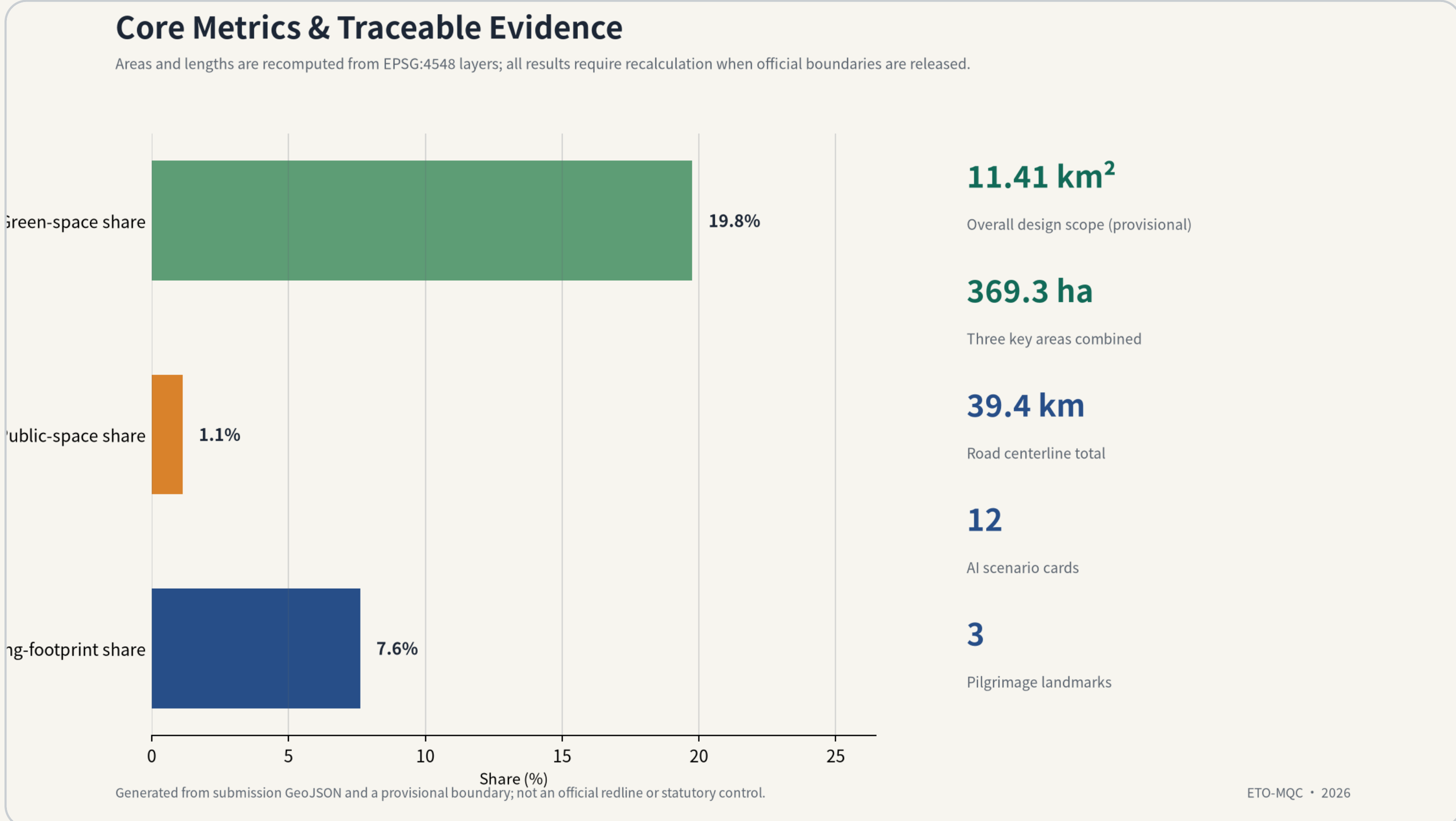

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Sources and rights | Bilingual equivalence | Hashes and four gates

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